AI Opportunity Hub Reference Architecture

Client-ready architecture overview for solution evaluation, implementation planning, and reuse.

1. Solution Overview

AI Opportunity Hub is a reference solution that helps organizations capture ideas, evaluate technical feasibility, generate architecture packages, and prepare implementation guidance for AI initiatives.

2. Logical Layers

- Presentation Layer: React + Vite experience for idea intake and review
- Application Layer: FastAPI endpoints for validation, orchestration, and package generation
- Intelligence Layer: context evaluation and architecture decision logic
- Data Layer: local persistence, document storage, and artifact management
- Integration Layer: deployment readiness for Azure services, authentication, and future enterprise connectors

3. Core Runtime Flow

User	Frontend	Idea capture
API	FastAPI	Validation + orchestration
Context Engine	Domain Logic	Business/technical assessment
Architecture Package	Output	Components, risks, deployment steps

4. Reference Deployment Model

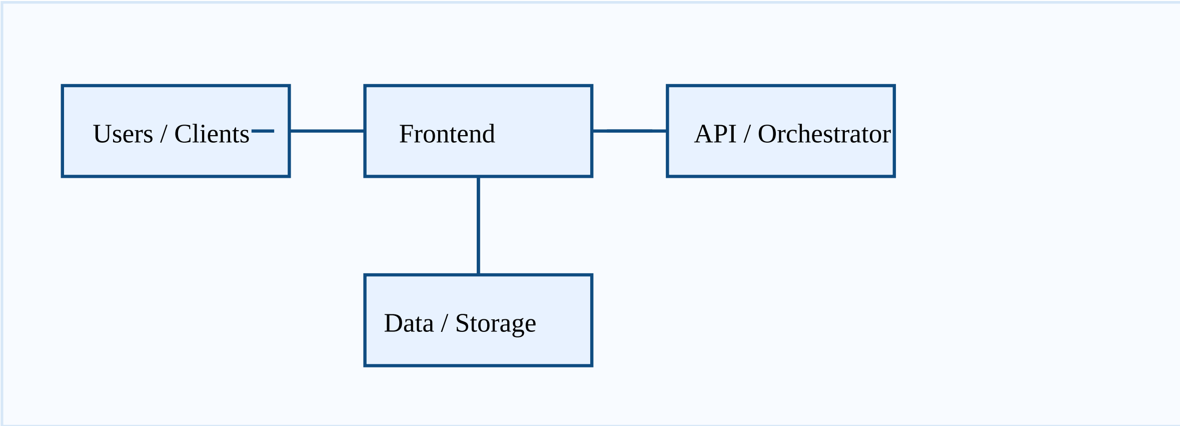
The architecture supports a local MVP deployment and a cloud-aligned Azure deployment path. The current implementation is optimized for rapid demonstration, while the structure is prepared for enterprise evolution.

5. Recommended Reuse Scenarios

- Use the frontend and API layout as a foundation for a custom innovation portal
- Reuse the validation workflow for internal AI opportunity review programs
- Adapt the storage abstraction to enterprise databases and document services
- Replace demo authentication with Microsoft Entra ID for production rollout

6. Architecture Summary

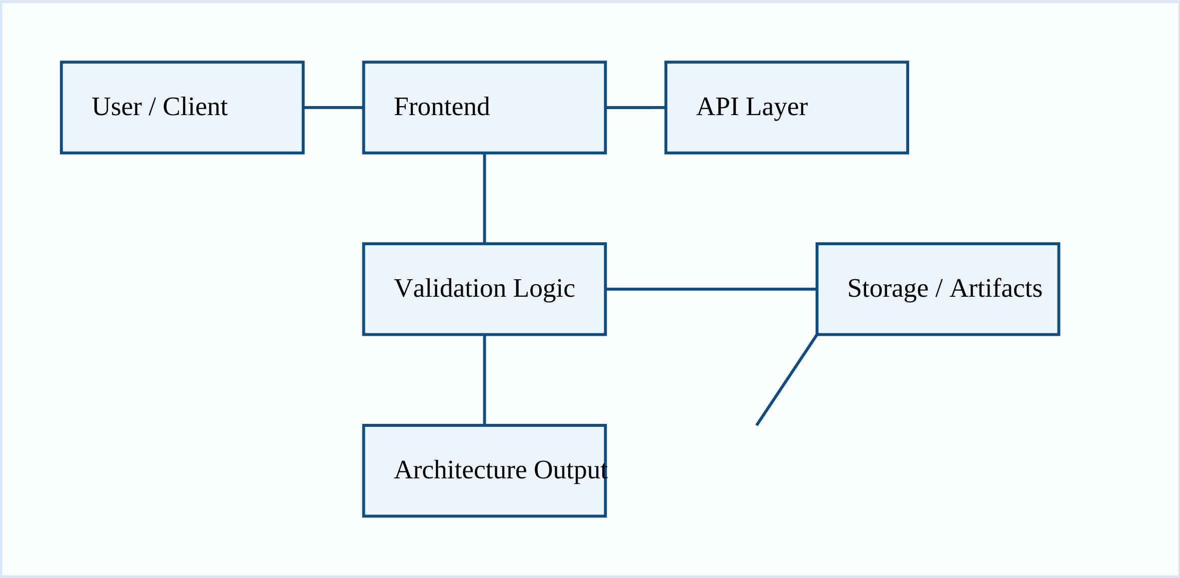
A visual summary of the solution follows.



This document is intended as a client-facing reference for architecture understanding, reuse, and implementation planning.

Tab 1 - System Flow Diagram

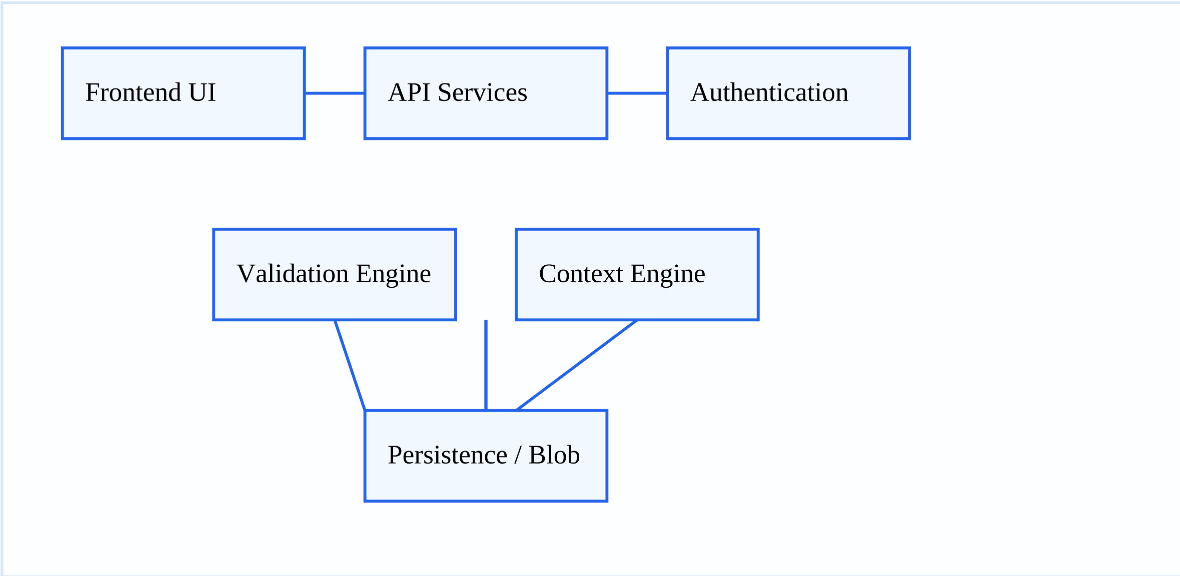
This view illustrates how a request moves through the platform from user interaction to generated architecture output.



Flow summary: capture idea → validate feasibility → generate architecture package → persist outputs and share results.

Tab 2 - Component Diagram

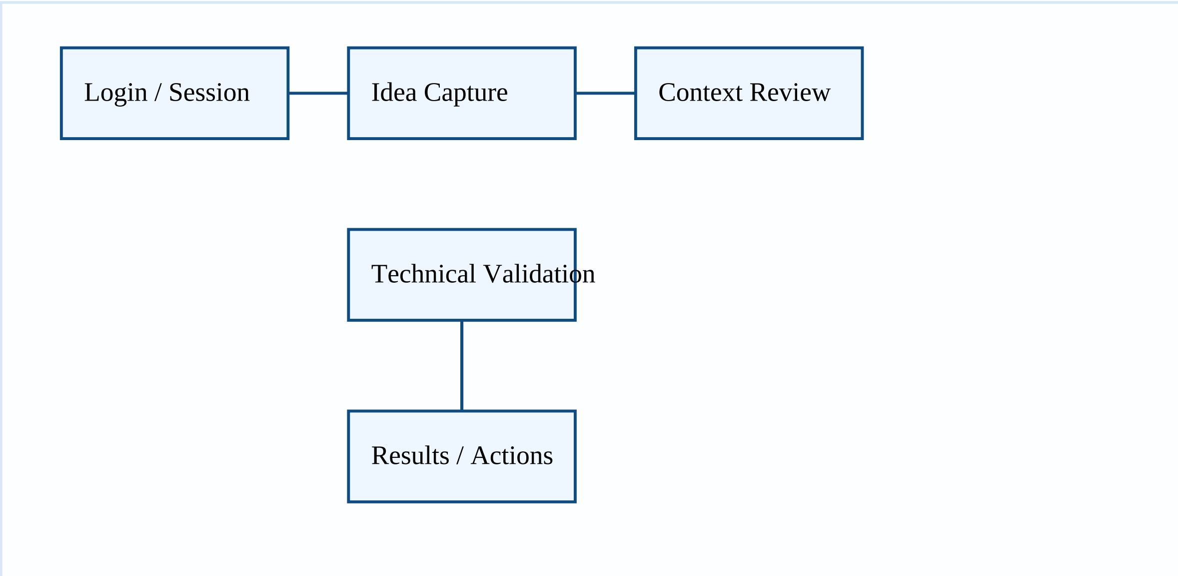
This view shows the main software components and their interaction boundaries for implementation and extension.



Component summary: UI, API, validation, context, storage and identity layers can evolve independently for future enterprise deployments.

Tab 3 - Frontend Flow Diagram

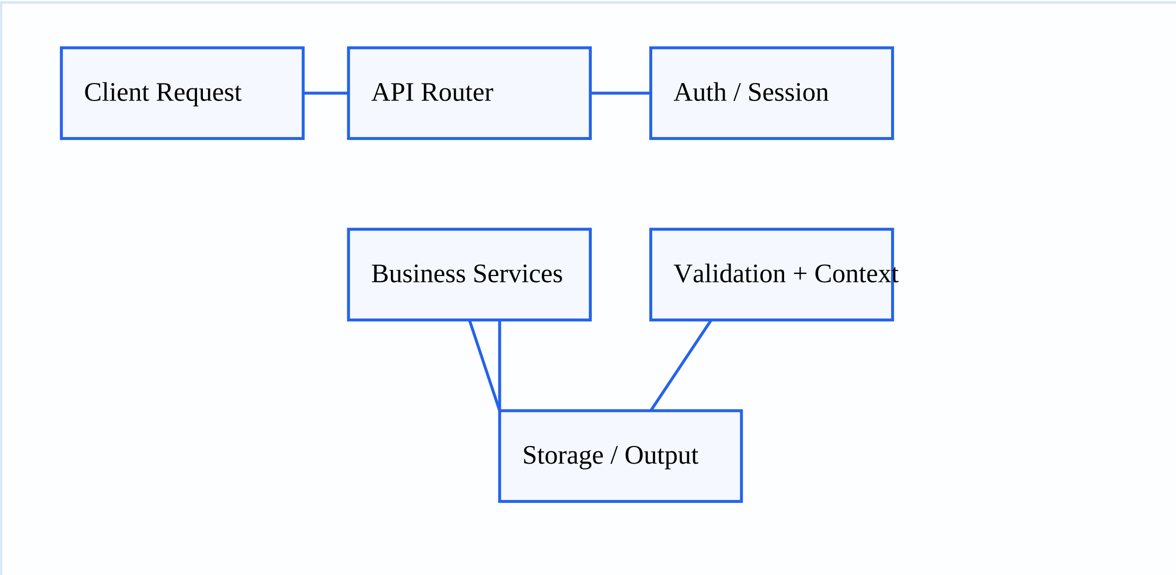
This view shows how the user interface drives the ideation and review experience end-to-end.



Frontend flow: session → capture idea → review context → validate and present guidance.

Tab 4 - Backend and API Layer Flow

This view describes the backend orchestration and API interactions behind the experience.



Backend flow: route request → authenticate → invoke services → persist artifacts and return response.